心脏的神经场：项目指南

第4组 · Neural Fields for Continuous Cardiac Cine-MRI · 帮助每个人理解整个项目，并能用自己的话讲清任何部分

> 老师会自己点名谁来讲、谁来回答。本指南的目的是让每个人理解项目的**意义**，而不是背数字。蓝色框里是可以直接说出口的简短英文句子。

1. 三句话讲完整个故事

MRI把跳动的心脏拍成一块一块的：几张很厚的切片和几十帧画面，中间的信息都丢失了。我们训练一个小神经网络，让它成为这颗心脏的“公式”：给它空间中的一个点和一个时刻，它告诉你那里看到的是什么。有了这个公式，就能得到任意位置、任意时刻的心脏，构建会跳动的三维模型，并计算心脏泵血的能力。

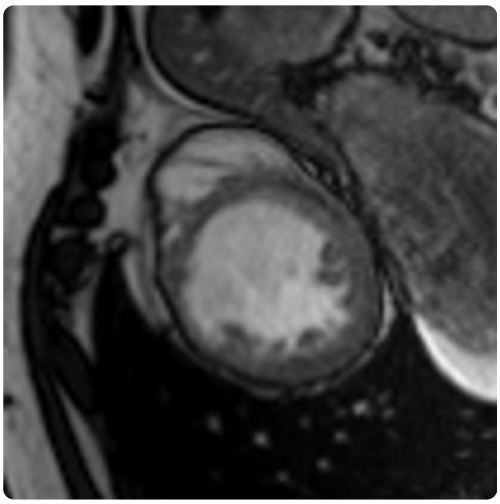
> *“We turn one heartbeat from a grid of pixels into one continuous function that can be asked about any point and any moment.”*

2. 什么是心脏电影MRI，问题在哪里

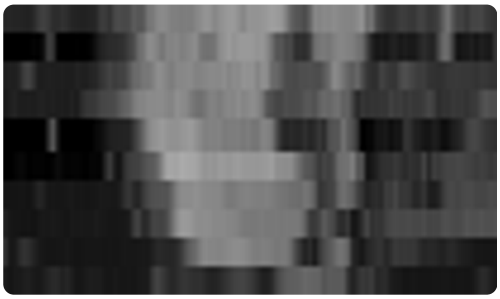
心脏电影MRI（cine-MRI）就是心脏的“视频”。沿着心脏长轴的垂直方向把心脏切成一片一片，像切面包一样：我们的病人有10个切片。每个切片拍摄一次心跳的大约30帧。

为什么这么粗？每个切片都要在病人屏住呼吸约10秒的时候拍摄。切片和帧越多，检查越长，屏气次数越多，病人承受不了。所以只能折中：切片内像素是1.6毫米（很清晰），切片之间却隔10毫米（空的）。帧与帧之间的运动也丢失了。

医生需要什么。最重要的数字是**射血分数**：心室每跳一次射出血液的比例。左心室正常约为50–70%。要计算它，需要心室在最舒张和最收缩时的体积。厚切片让体积变成“台阶状”，不够准确。



一个切片：心脏横截面，亮的是血液，暗的环是肌肉。



横穿10个切片的截面：心脏由10毫米的“砖块”组成。

3. 核心想法：表格还是公式

类比。普通图片是一张数字表格：每个格子存一个亮度。放大就看到方块，格子之间什么也没有。矢量图（比如屏幕上的字体）存的不是格子，而是形状的公式，所以可以无限放大。

我们为跳动的心脏做“矢量图”，但公式不是手写的，而是由神经网络学出来的。网络输入4个数——**x, y, z**（在哪里，毫米）和**t**（什么时候，心跳中的哪个时刻）——输出1个数：该点的亮度。训练结束后，**网络本身就是这次扫描**。这类网络叫**神经场**（neural fields，隐式神经表示）。

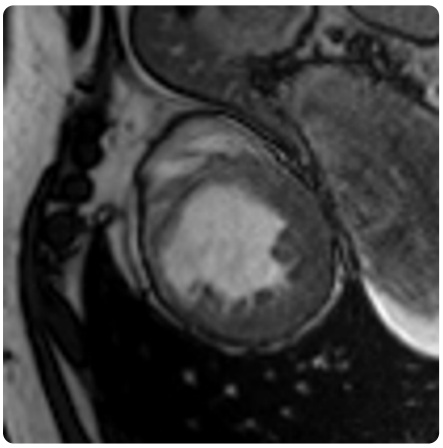


它怎么学习。我们取扫描中所有测量到的点（几百万对“坐标 → 亮度”），分批给网络看，并调整它的权重，使回答接近真实值。20000步这样的训练，在普通笔记本上约25分钟。

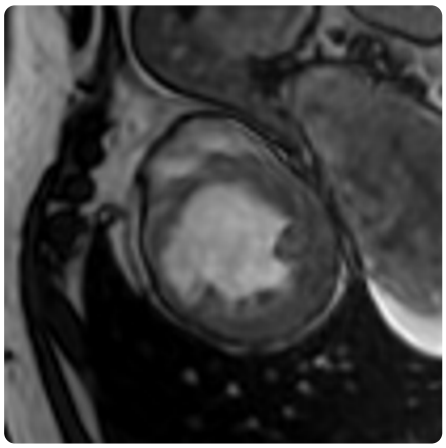
> *“After training, the network is the scan: four numbers in, one brightness out.”*

4. 为什么用正弦（SIREN）而不是普通网络

普通网络（ReLU）用直线段拼出答案。它很容易画出平滑的形状，但细节和清晰的边缘会被“抹平”。**SIREN**则用正弦来组合答案——不同频率的波，就像声音由音符组成。波既能画平滑的部分，也能画清晰的边缘。



真实图像（网络没见过的帧）



SIREN：清晰的心脏



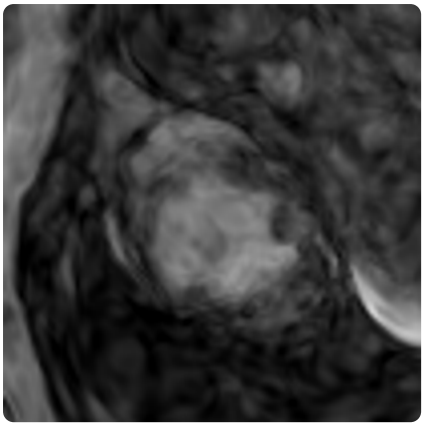
同样大小的ReLU：模糊的一团

两个网络的权重数量完全相同，训练也相同。唯一的区别是神经元的函数，而质量相差6.6 dB（误差约小4.5倍）。

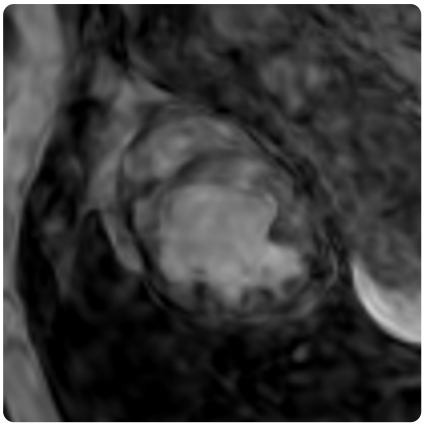
5. 我们的发现：让正弦在时间和切片方向上变慢

问题。$\omega_0 = 30$的正弦变化非常“快”。网络能完美记住看到的帧，但在帧与帧之间开始振荡，就像钉子之间绷紧的琴弦。在没见过的帧上得到的是垃圾。

解决办法。把时间输入网络之前，先乘以一个小数（0.1）。对网络来说时间好像被“拉长”了，于是它在时间上变化得更平滑。切片编号也一样：$z \times 0.25$。这是我们自己的贡献：原始SIREN论文里没有这一点，而对四维心脏来说这是决定性的改动。



切片之间不缩放：噪声



缩放 z × 0.25：平滑

+6.6 dB

仅仅因为时间输入方式的改变：22.7 → 29.3 dB

> *“Scaling one input slows the sines along that axis only, so the network stops oscillating between the frames it saw.”*

6. 我们如何诚实地检验

考试的类比。如果用网络训练过的帧来检验，就像考试题提前发给了学生。所以我们**隐藏**一部分扫描：例如网络只看到每第四帧（30帧中的9帧），之后让它重建另外21帧，再与真实值比较。

和谁比较。（1）**线性插值**——猜测缺失帧最简单的方法：把相邻两个已知帧混合，好像心脏沿直线运动。这是很强的对手。（2）同样大小的普通ReLU网络。（3）带Fourier features的ReLU网络——普通网络的强化版本。

如何避免针对测试调参。设置（例如时间缩放0.1）在单独的**验证**帧（第12和第20帧）上选择：这些帧从训练中移除，只用于选择。测试帧不参与选择。SIREN用三个随机种子重复。最重要的是，最后我们用已经固定的设置在**14个新病人**（三个疾病组）上运行了整个流程，这些病人从未用于任何选择。

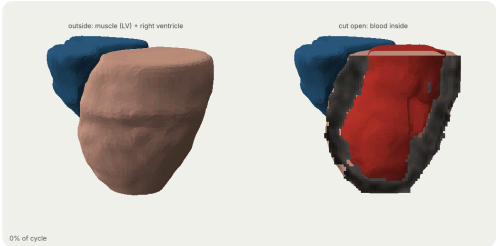
用什么衡量。PSNR，单位是分贝（dB）：越大越接近真实；+10 dB表示误差小10倍。只在心脏内部计算（根据医生的标注），因为黑色背景很容易重建，会抬高分数。

7. 结果，用简单的话说

- **正弦确实必要：**SIREN比同样大小的普通网络好6.6 dB（在14个新病人上平均好7.4 dB，全部14个都更好），比带Fourier features的ReLU好2.3 dB。三个不同种子的结果只相差0.04 dB。
- **与插值几乎持平：**28.9对29.2 dB。原因很清楚：一次心跳30帧已经很密，相邻帧之间心脏几乎沿直线运动，而直线正是插值。
- **在心脏运动最快的时候**（快速收缩期），网络**比插值好**：那里的运动是弯曲的，直线跟不上。
- **两者在不同地方出错，**所以“网络 + 插值”的平均比两者都好：29.7 dB，21帧中18帧更好。
- **14个新病人：**混合方法在**全部14个**病人上都比插值好（+0.37 dB，$p = 0.0001$）。单独的SIREN平均比插值落后0.6 dB。这是最主要的诚实结论：网络单独无法超过插值，但两者结合稳定更好。
- **紧凑：**整个心跳的心脏区域只需0.8 MB权重——数字比体素少16倍；整个扫描少86倍，质量只损失一点。
- **切片之间**相隔20毫米时，网络和插值都猜不出来：心脏在20毫米内形状真的会变。这是数据的限制。

8. 如何从网络得到跳动的三维心脏

在很密的网格上询问网络：每1.6毫米一个切片，而不是10毫米（比扫描仪密6倍），时刻可以任选。MRI上血液是亮的，肌肉是暗的，所以用亮度阈值找到血液所在的位置。**marching cubes**算法用三角形表面把这个区域“包起来”。对30个时刻重复——就得到会跳动的三维模型（这就是四维）。



左边是心脏外观，右边是剖开的：剖面上是网络自己画出的图像，里面是红色的血液。

验证模型不只是好看：在数据给定的舒张末期和收缩末期帧上，从模型计算体积和射血分数。左心室：我们42.7%，医生43.6%。在14个新病人上，与医生轮廓的重合度Dice为0.87，与同样阈值在原始扫描上的结果相同，说明网络没有破坏形状；射血分数平均误差4.6个百分点。体积偏小，因为阈值没有计入心室内部的小肌肉。较薄的右心室用简单阈值无法可靠分割——这是我们分割方法的局限。

9. 这种模型在现实中有什么用

我们的项目是研究原型，不是可以直接使用的医疗设备。但“把扫描当作连续函数”这个思想已经在被使用和积极发展。

领域	神经场带来什么
心脏病学	在厚切片之间查看心脏，计算没有“台阶”的体积和射血分数；测量心壁如何运动和变形——这需要导数，而光滑函数的导数可以精确计算。
更快的MRI	研究人员用这类网络从严重欠采样的数据中重建电影MRI：扫描更快，屏气次数更少。
图像配准	MRI、CT和超声在不同网格上拍摄。以毫米为单位的连续函数可以把它们放进同一坐标系比较——这也是我们项目的动机。
病人心脏模型	光滑的三维/四维模型是“数字孪生”的基础：血流模拟、手术规划、3D打印。
存储和传输	一个小的权重文件代替庞大的体素数组；可以按任意分辨率查询。
医学以外	同样的原理用于NeRF：从普通照片得到三维场景（用手机视频生成3D模型的应用）、神经网络图像和视频压缩、图形和游戏中的三维形状、物理仿真。

> *“The same idea is used for 3D scenes from photos (NeRF), for faster MRI reconstruction and for patient-specific heart models.”*

10. 诚实的局限

- 单独的网络平均比插值差0.6 dB；在快速阶段和混合方法中胜出。
- 切片之间20毫米的空隙，任何方法都无法恢复。
- 三维模型借助医生标注、用简单的亮度阈值构建；右心室不可靠。
- 我们尝试过让网络单独超过插值：样条插值比线性插值更差，只有考虑运动的方法更好（光流 +0.3 dB）。简单的运动神经网络没有成功——这是下一步，同样的还有从许多病人学到的先验。

不知道时怎么回答：“Good question. We did not test that; what we measured is …”——然后说出最接近的结果。

附录A. 值得记住的数字

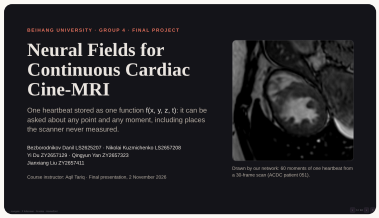
Scan of patient 051	224 × 256 px of 1.64 mm, 10 slices × 10 mm, 30 frames = 17.2 M numbers 224 × 256像素（1.64毫米），10个切片×10毫米，30帧 = 1720万个数
Network (SIREN)	4 layers × 256 sine neurons, $\omega_0 = 30$, 198,913 weights ≈ 0.8 MB 4层×256个正弦神经元，$\omega_0 = 30$，198,913个权重 ≈ 0.8 MB
Training	Adam, 20,000 steps, ≈ 25 min on a laptop CPU Adam，20000步，笔记本CPU约25分钟
Time scale	raw t 22.7 dB → scaled 29.3 (+6.6); 0.1 chosen on validation frames 直接输入t 22.7 dB → 缩放后29.3（+6.6）；0.1在验证帧上选出
SIREN vs ReLU	28.9 vs 22.3 dB (+6.6); vs ReLU + Fourier features 26.6 (+2.3); 3 seeds 28.91 ± 0.04 28.9对22.3 dB（+6.6）；对ReLU + Fourier features 26.6（+2.3）；3个种子28.91 ± 0.04
vs interpolation	patient 051, every 4th kept: 28.9 vs 29.2; blend 29.7 (wins 18 of 21) 051号病人，每第四帧保留：28.9对29.2；混合29.7（21帧中18帧更好）
14 new patients	blend > interpolation on 14 / 14 (+0.37 dB, p = 0.0001); SIREN > ReLU 14 / 14 (+7.4); SIREN alone −0.57 混合 > 插值，14个中14个（+0.37 dB，p = 0.0001）；SIREN > ReLU 14个中14个（+7.4）；单独SIREN −0.57
Compression	heart region 16× (28.9 dB); whole scan 86× (27.5 dB) 心脏区域16倍（28.9 dB）；整个扫描86倍（27.5 dB）
Hidden slices (20 mm gap)	all ≈ 20 dB; SIREN with z × 0.25: 20.09 vs interpolation 20.01 都约20 dB；z×0.25的SIREN 20.09对插值20.01
Ejection fraction	LV 42.7 % (experts 43.6 %) at ED/ES; 14 patients: LV Dice 0.87, EF error 4.6 points; RV unreliable 左心室42.7%（专家43.6%），ED/ES帧；14个病人：左心室Dice 0.87，EF误差4.6个百分点；右心室不可靠

附录B. 术语

Neural field / implicit neural representation	a network that IS the signal: you give it coordinates, it gives the value there; it can be asked anywhere 网络本身就是信号：输入坐标，输出该处的值；可以在任何位置询问
SIREN	a network whose neurons compute $\sin(\omega_0(Wx + b))$; keeps sharp detail 神经元计算$\sin(\omega_0(Wx + b))$的网络；能保留清晰细节
ReLU network	the usual network with max(0, x); learns smooth shapes first and blurs detail 普通的max(0, x)网络；先学平滑形状，会模糊细节
$\omega_0 = 30$	how fast the sines oscillate; from the SIREN paper 正弦振荡的快慢；来自SIREN论文

Time / z scale	we multiply t (or z) by 0.1 (or 0.25) inside the model, so the field changes slower along that axis 在模型内把t（或z）乘以0.1（或0.25），让场沿该轴变化更慢
Linear interpolation	a straight line between the two known neighbours: (frame 4 + frame 8) mixed by distance 两个已知相邻帧之间的直线：按距离混合第4帧和第8帧
PSNR, dB	how close the copy is to the truth; higher is better; +10 dB = 10× smaller squared error 副本与真值的接近程度；越高越好；+10 dB = 平方误差小10倍
SSIM	structural similarity of two images, 0 to 1; higher is better 两幅图像的结构相似度，0到1；越高越好
Hidden data	slices or frames removed from training and used only for scoring 从训练中移除、只用于打分的切片或帧
Voxel, NIfTI	a 3D pixel; NIfTI is the medical file format (.nii) storing the 4D scan 三维像素；NIfTI是存储4D扫描的医学文件格式（.nii）
End-diastole / end-systole	the heart fully relaxed (largest) / fully contracted (smallest) 心脏完全舒张（最大）/ 完全收缩（最小）
Ejection fraction (EF)	share of blood pushed out per beat: (largest − smallest volume) / largest 每次心跳射出的血液比例：（最大−最小体积）/ 最大体积
Marching cubes	algorithm that turns a 3D grid into a triangle surface at a chosen level 把三维网格在给定阈值处转换成三角形表面的算法

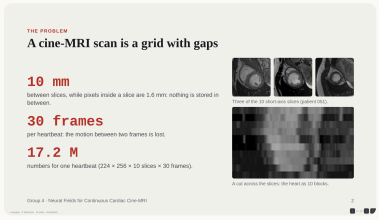
附录C. 每页幻灯片讲什么（英文 + 翻译）



1 · Title

Good morning. We are Group 4, and our project is Neural Fields for Continuous Cardiac Cine-MRI. The beating heart on the right is not the scan. It is drawn by our trained network: 60 moments of one heartbeat, although the scanner recorded only 30. Today we show how we store a whole heartbeat as one function, how we tested it, and what came out of it.

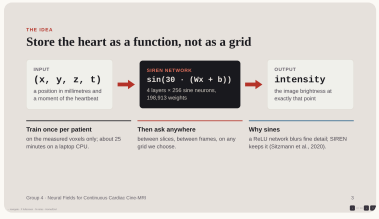
> 早上好。我们是第4组，项目是“用于连续心脏电影MRI的神经场”。右边跳动的心脏不是扫描本身，而是我们训练好的网络画出来的：一次心跳的60个时刻，而扫描仪只记录了30个。今天我们介绍如何把一次完整的心跳存成一个函数、我们如何检验它，以及得到了什么结果。



2 · The problem

A cardiac MRI scan is stored as a grid. Inside a slice, one pixel is 1.6 millimetres, but the slices are 10 millimetres apart, so between them nothing is stored. One heartbeat has only 30 frames, so the motion between frames is lost. And one heartbeat takes 17 million numbers. In the cut across the slices, the heart looks like ten blocks. The real heart is smooth. Our question: can we store it as something continuous?

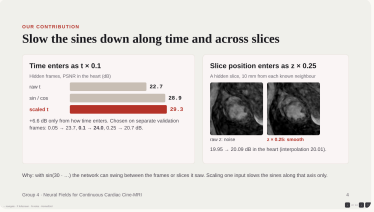
> 心脏MRI扫描是以网格形式存储的。在一个切片内，像素是1.6毫米，但切片之间相隔10毫米，中间什么也没有存。一次心跳只有30帧，帧与帧之间的运动就丢失了。而一次心跳需要1700万个数。从横穿切片的截面看，心脏像十个方块。真实的心脏是光滑的。我们的问题是：能不能把它存成连续的东西？



3 · The idea

Our idea: store the heart as a function, not as a grid. A small network takes four numbers, the position x, y, z in millimetres and the moment t, and returns the brightness at that point. It is a SIREN: every neuron computes a sine. Four layers of 256 neurons, about 199 thousand weights. We train it once per patient, about 25 minutes on a laptop, and then we can ask it anywhere: between slices and between frames. We use sines because a normal ReLU network blurs fine detail.

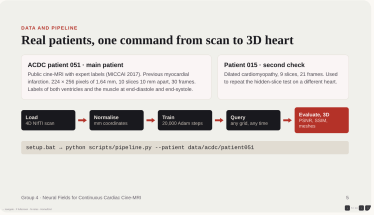
> 我们的想法：把心脏存成一个函数，而不是网格。一个小网络输入四个数——以毫米为单位的位置x、y、z和时刻t——输出该点的亮度。它是SIREN：每个神经元计算一个正弦。四层，每层256个神经元，约19.9万个权重。每个病人训练一次，在笔记本上约25分钟，之后可以在任何地方提问：切片之间、帧之间。用正弦是因为普通的ReLU网络会把细节变模糊。



4 · Our contribution

The plain SIREN was not enough. Our own contribution is two simple changes. Left, time: with raw t the network learns the frames it saw and swings between them, only 22.7 dB on hidden frames. If we multiply t by a small number inside the model, we get 29.3 dB, plus 6.6 dB. The best value, 0.1, was chosen on separate validation frames, not on the test. Right, the same problem across slices: with raw z the hidden slice is full of noise; z times 0.25 makes it smooth. Scaling one input slows the sines along that direction only.

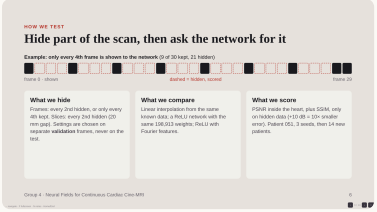
> 普通的SIREN不够。我们自己的贡献是两个简单的改动。左边是时间：直接输入t时，网络记住看到的帧并在它们之间振荡，隐藏帧只有22.7 dB。如果在模型内把t乘以一个小数，得到29.3 dB，提高6.6 dB。最佳值0.1是在单独的验证帧上选出的，而不是在测试帧上。右边是切片方向的同样问题：直接输入z时，隐藏切片充满噪声；z乘以0.25后变得平滑。缩放某一个输入，只会让正弦沿这个方向变慢。



5 · Data and pipeline

Our data is real: the public ACDC dataset from the MICCAI 2017 challenge, with labels from experts. Our main patient, 051, had a heart attack: 10 slices, 30 frames, and labels of both ventricles at two moments. Patient 015, with a different disease, is our second check. The pipeline has five steps: load the scan, turn voxels into coordinates in millimetres, train, ask the network on any grid, then evaluate and build the 3D heart. One setup file installs everything, and one command runs the whole study.

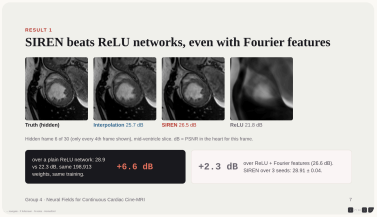
> 我们的数据是真实的：来自MICCAI 2017挑战赛的公开ACDC数据集，带有专家标注。主要病人051曾患心肌梗死：10个切片、30帧，在两个时刻有两个心室的标注。病人015患另一种疾病，是我们的第二次检验。流程有五步：读取扫描、把体素变成毫米坐标、训练、在任意网格上询问网络、最后评估并构建三维心脏。一个安装文件装好一切，一条命令运行整个研究。



6 · How we test

To test honestly, we hide part of the scan and ask the network for it. Here the network sees only every fourth frame, the dark ones; the 21 dashed frames are hidden and scored. We also hide every second frame, or every second slice. We compare with linear interpolation between the known neighbours, with a ReLU network of the same size, and with a ReLU network with Fourier features. We score only hidden data, with PSNR inside the heart; plus 10 dB means ten times smaller error. Settings such as the time scale are chosen on separate validation frames, never on the test frames, and at the end we repeat everything on 14 new patients.

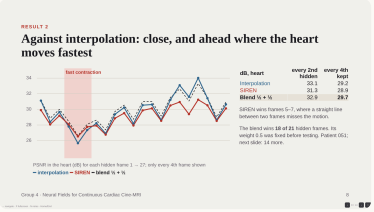
> 为了诚实地检验，我们隐藏扫描的一部分，再让网络把它恢复出来。这里网络只看到每第四帧（深色的）；21个虚线帧被隐藏并用于评分。我们也隐藏每第二帧或每第二个切片。我们与已知相邻帧之间的线性插值、同样大小的ReLU网络以及带Fourier features的ReLU网络比较。只在隐藏数据上评分，用心脏内的PSNR；提高10 dB表示误差小10倍。时间缩放等设置在单独的验证帧上选择，从不在测试帧上，最后我们在14个新病人上重复全部实验。



7 · Result 1: SIREN vs ReLU networks

Result one. This is frame 6, which the network never saw. Interpolation gives 25.7 dB, SIREN 26.5, and the ReLU network only a blurry shape, 21.8. Over all 21 hidden frames SIREN is 6.6 dB better than ReLU, with the same number of weights. A stronger baseline, ReLU with Fourier features, closes part of the gap, 26.6 dB, but SIREN is still 2.3 dB better; and repeating SIREN with three random seeds changes the result by only 0.04 dB. This is why we use sines.

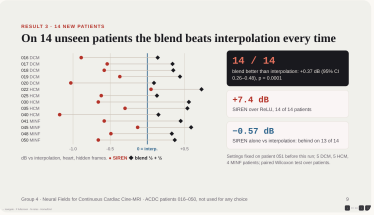
> 第一个结果。这是网络从未见过的第6帧。插值得到25.7 dB，SIREN 26.5，而ReLU网络只是一团模糊，21.8。在全部21个隐藏帧上，SIREN在相同权重数量下比ReLU好6.6 dB。更强的基线——带Fourier features的ReLU——缩小了一部分差距，达到26.6 dB，但SIREN仍然好2.3 dB；用三个不同随机种子重复SIREN，结果只相差0.04 dB。这就是我们使用正弦的原因。



8 · Result 2: vs interpolation

Result two, the strong baseline: interpolation. To be honest, on average it is still a little ahead: 29.2 against our 28.9 dB, a gap of 0.3 dB. But look at the shaded part: when the heart contracts fastest, a straight line between two frames misses the motion, and SIREN wins. The two methods make different mistakes, so we averaged them with a fixed weight of one half. This blend beats interpolation: 29.7 dB, better on 18 of 21 frames.

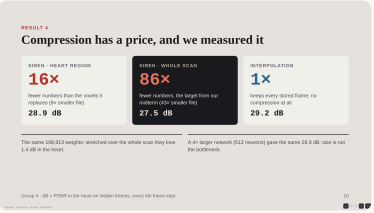
> 第二个结果，强有力的基线：插值。老实说，平均来看它仍然略好：29.2对我们的28.9 dB，相差0.3 dB。但请看阴影部分：心脏收缩最快时，两帧之间的直线跟不上运动，SIREN更好。两种方法的错误出现在不同地方，所以我们用事先固定的二分之一权重把它们平均。这个混合超过了插值：29.7 dB，在21帧中的18帧更好。



9 · Result 3: 14 new patients

One patient is not enough, so we tested 14 new patients from three disease groups, with all settings fixed in advance on patient 051. The picture is the same. On every patient the network alone is about half a dB behind interpolation, but the blend of the two is better than interpolation on all 14 patients: plus 0.37 dB on average, and the paired test gives p of 0.0001. And sines beat ReLU on every patient, by 7.4 dB on average.

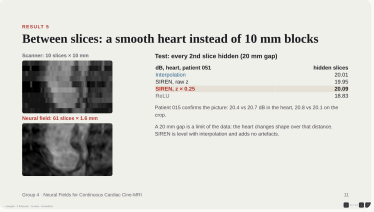
> 一个病人不够，所以我们在三个疾病组的14个新病人上做了测试，所有设置都事先在051号病人上固定。结果一样。每个病人上，单独的网络比插值落后大约半个分贝，但两者的混合在全部14个病人上都比插值好：平均高0.37 dB，配对检验p = 0.0001。而且在每个病人上正弦网络都超过ReLU，平均高7.4 dB。



10 · Result 4: compression

Result three: compression. At the midterm we promised about 86 times fewer numbers. The same network can store only the heart region, 16 times fewer numbers, at 28.9 dB, or the whole scan, 86 times fewer, at 27.5 dB. Interpolation compresses nothing. A four times bigger network did not help, so size is not our limit.

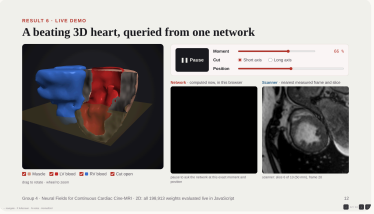
> 第三个结果：压缩。在期中我们承诺大约少86倍的数字。同一个网络可以只存心脏区域，少16倍，28.9 dB；也可以存整个扫描，少86倍，27.5 dB。插值完全不压缩。大四倍的网络没有帮助，所以网络大小不是我们的瓶颈。



11 · Result 5: between slices

Between slices. The scanner gives ten blocks; our network, asked every 1.6 millimetres, gives a smooth heart from base to tip. When we hid every second slice, a 20 millimetre gap, no method could really fill it: all are near 20 dB. With scaled z, SIREN is level with interpolation and has no noise, and the second patient shows the same.

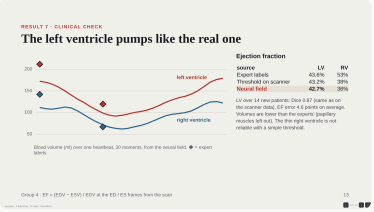
> 切片之间。扫描仪给出十个方块；我们的网络每1.6毫米询问一次，给出从心底到心尖的光滑心脏。当我们隐藏每隔一个切片，即20毫米的间隔时，没有任何方法能真正填补：都在20 dB左右。缩放z之后，SIREN与插值持平且没有噪声，第二个病人也一样。



12 · Result 6: 3D heart, live demo

This is our main result: one network trained on all frames, asked on a grid six times finer than the scanner. We build surfaces where the brightness changes from blood to muscle. On the left, the heart from outside; on the right, cut open: the cut shows the network's own image of the wall, and the red blood inside contracts. Now a short live demo. [in the HTML presentation: rotate the heart, press Pause — the network computes the slice live at this moment; move the time slider between two frames, switch to the long axis — about 40 seconds]

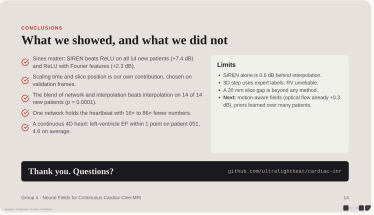
> 这是我们的主要结果：一个在全部帧上训练的网络，在比扫描仪细六倍的网格上询问。在亮度从血液变为肌肉的地方构建表面。左边是心脏外观；右边是剖开的：剖面显示网络自己生成的心壁图像，里面红色的血液在收缩。现在是简短的现场演示。[在HTML演示中：旋转心脏，按Pause——网络实时计算这一时刻的切片；把时间滑块移到两帧之间，切换到长轴——约40秒]



13 · Result 7: clinical check

Is it correct? Doctors judge a heart by its ejection fraction. At the end-diastolic and end-systolic frames given with the scan, our 4D model gives 42.7 percent for the left ventricle, the experts 43.6. Over the 14 new patients the left ventricle overlaps the expert outline with Dice 0.87, the same as the threshold on the original scan, so the network does not spoil the shape, and the ejection fraction is off by 4.6 points on average. The thin right ventricle is not reliable with a simple threshold: this is a limit of our segmentation, not of the network.

> 它准确吗？医生用射血分数来评价心脏。在数据给定的舒张末期和收缩末期帧上，我们的四维模型得到左心室42.7%，专家为43.6%。在14个新病人上，左心室与专家轮廓的重合度Dice为0.87，与在原始扫描上用同样阈值的结果相同，说明网络没有破坏形状；射血分数平均相差4.6个百分点。较薄的右心室用简单阈值无法可靠分割：这是我们分割方法的局限，而不是网络的局限。



14 · Conclusions

To sum up: sines matter, on every patient; scaling time and slice position is our own contribution; the blend of network and interpolation beats interpolation on all 14 new patients; and from one network we get a 4D heart whose left ventricle agrees with the experts. Our limits: the network alone is still about half a dB behind interpolation, and the 3D step uses expert labels. Next we want fields that model motion, since classical optical flow already beats linear interpolation, and priors learned over many patients. Thank you, we are happy to answer your questions.

> 总结：正弦很重要，在每个病人上都是如此；缩放时间和切片位置是我们自己的贡献；网络与插值的混合在全部14个新病人上都超过插值；从一个网络得到的四维心脏，其左心室与专家一致。局限：单独的网络仍比插值落后大约半个分贝，三维步骤使用了专家标注。下一步我们希望做能建模运动的神经场（经典光流已经超过线性插值），以及从许多病人身上学到的先验。谢谢，欢迎提问。

附录D. 可能的问题与回答

1. Interpolation is better on average. Why use a network at all?

Interpolation needs every stored frame; our network is one function with 16 to 86 times fewer numbers, it can be asked anywhere, and it is differentiable, so motion and surfaces come directly from it. Where motion is fastest it already wins, and the blend of both beats interpolation on all 14 new patients.

> 插值需要所有存储的帧；我们的网络是一个函数，数字少16到86倍，可以在任何地方询问，而且可微，运动和表面可以直接从中得到。在运动最快的地方它已经更好，两者的混合在全部14个新病人上都超过了插值。

2. Is this not just overfitting?

Yes, on purpose: one network per patient is the standard setting for neural fields. That is exactly why we never score the training data, only slices and frames that were hidden.

> 是的，这是有意的：每个病人一个网络是神经场的标准设定。正因如此，我们从不在训练数据上打分，只在隐藏的切片和帧上打分。

3. Why sines? Why $\omega_0 = 30$?

ReLU networks learn smooth, low-frequency functions first and blur detail; sines can represent sharp edges. $\omega_0 = 30$ and the initialisation are from the SIREN paper. Instead of changing ω_0 we scale t and z, which acts like a separate ω_0 per axis.

> ReLU网络先学习平滑的低频函数，会模糊细节；正弦可以表示清晰的边缘。$\omega_0 = 30$和初始化来自SIREN论文。我们不改ω_0，而是缩放t和z，相当于每个轴有各自的ω_0。

4. What exactly is your own contribution?

Per-axis scaling of time and slice position for a 4D heart, with a sweep; an honest protocol on hidden data against interpolation and a same-size ReLU; the fixed blend; the measured compression–quality trade-off; the 4D heart with an ejection-fraction check; and an open one-command pipeline.

> 为四维心脏按轴缩放时间和切片位置并做了参数扫描；在隐藏数据上与插值和同样大小的ReLU进行诚实比较；固定权重的混合；测量了压缩与质量的权衡；带射血分数验证的四维心脏；以及一条命令运行的开源流程。

5. Why measure PSNR only in the heart?

Most of the image is dark background that every method reconstructs easily, and it inflates PSNR: over the whole image interpolation gets 38.6 dB but only 29.2 in the heart. The heart is what moves and what matters.

> 图像大部分是黑色背景，任何方法都很容易重建，它会抬高PSNR：在整幅图像上插值是38.6 dB，在心脏内只有29.2。真正运动、真正重要的是心脏。

6. Only one patient. Does it generalise?

The method is per patient: every patient gets its own network. To check generalisation we fixed all settings on patient 051 and ran the same protocol on 14 new patients from three disease groups: SIREN beats ReLU on 14 of 14, the blend beats interpolation on 14 of 14 ($p = 0.0001$), and SIREN alone is about 0.6 dB behind interpolation, the same picture as on patient 051.

> 该方法是针对单个病人的：每个病人有自己的网络。为了检验泛化，我们在051号病人上固定所有设置，再在三个疾病组的14个新病人上运行同样的流程：SIREN在14个病人中全部超过ReLU，混合方法在14个病人中全部超过插值（$p = 0.0001$），单独的SIREN比插值约落后0.6 dB，与051号病人上的结果一致。

7. How long does training take? Would a GPU change the results?

About 25 minutes for 20,000 steps on an Intel Core Ultra 5 laptop CPU. A GPU gives the same results, only 10–50 times faster.

> 在Intel Core Ultra 5笔记本CPU上20000步约25分钟。GPU给出相同的结果，只是快10到50倍。

8. Is the network really smaller than the scan?

199 thousand 32-bit weights, about 0.8 MB. The heart region is 3.2 million 16-bit values, about 6.5 MB: 8 times smaller in bytes. The whole scan is 17.2 million values, about 34 MB: 43 times smaller in bytes, 86 times fewer numbers.

> 19.9万个32位权重，约0.8 MB。心脏区域是320万个16位数值，约6.5 MB：按字节小8倍。整个扫描是1720万个数值，约34 MB：按字节小43倍，数字少86倍。

9. Why did the periodic (sin/cos) time encoding not win?

It was close (28.9 vs 29.3 dB). With one recorded cycle the main problem is the speed of the field in time, not the link between the last and the first frame; scaling fixes the speed directly. Periodic encoding should matter more with several cycles.

> 它很接近（28.9对29.3 dB）。只记录一个周期时，主要问题是场在时间上的变化速度，而不是最后一帧与第一帧的衔接；缩放直接解决速度问题。多个周期时周期编码会更重要。

10. Why can the network beat interpolation during fast contraction?

Interpolation draws a straight line between two frames. When the wall moves fast and along a curve, the straight line misses it. The network is fitted to all frames at once and can follow a curved path in time.

> 插值在两帧之间画一条直线。当心壁快速地沿曲线运动时，直线就会偏离。网络同时拟合所有帧，可以沿时间上的曲线轨迹变化。

11. Why is the 20 mm hidden-slice test so poor for everyone?

Over 20 mm the heart really changes shape, so the information is not in the neighbouring slices. This is a limit of the data, not of the method; with scaled z the network is at least level with interpolation and without artefacts.

> 在20毫米内心脏的形状确实会改变，相邻切片中没有这些信息。这是数据的限制，不是方法的限制；缩放z后网络至少与插值持平且没有伪影。

12. How do you get a 3D surface from the network? Did you use the labels?

We ask the network on a fine grid, threshold blood against muscle and run marching cubes. Labels only say which bright region is which ventricle and set the threshold; the shape and motion come from the network.

> 我们在细网格上询问网络，用阈值区分血液和肌肉，然后运行marching cubes。标注只用来区分哪个亮区是哪个心室并确定阈值；形状和运动来自网络。

13. Why are your volumes lower than the experts' but the ejection fraction close?

Expert labels count the papillary muscles and trabeculae as part of the cavity; our brightness threshold leaves them out, so volumes are lower. The ejection fraction is a ratio, so for the left ventricle it stays close: 42.7 vs 43.6 % on patient 051 and 4.6 points on average over 14 patients. For the thin right ventricle a threshold is not reliable, which we list as a limitation.

> 专家把乳头肌和肌小梁算作心腔的一部分；我们的亮度阈值不包括它们，所以体积偏低。射血分数是比值，所以左心室的结果接近：051号病人为42.7%对43.6%，14个病人平均相差4.6个百分点。对较薄的右心室，阈值方法不可靠，我们把它列为局限。

14. Why a blend weight of 0.5? Is it tuned?

It was fixed before testing on purpose, so the test stays fair, and it was never changed: the same 0.5 is used for all 14 new patients, where the blend wins on every one. A weight chosen on validation frames might be slightly better; we did not tune it to keep the test clean.

> 它是事先固定的，这样测试保持公平，之后也从未改变：14个新病人都使用同样的0.5，混合方法在每个病人上都更好。在验证帧上选择的权重可能会稍好一点；为了保持测试干净，我们没有调它。

15. Could this be used in a hospital?

Not as it is: it is a research prototype, tested on 16 patients of one public dataset, with a label-assisted 3D step. Possible uses are viewing between slices, measuring motion from the derivatives of the field, compact storage, and aligning scans from different machines in one coordinate system.

> 目前还不行：这是在一个公开数据集的16个病人上测试的研究原型，三维步骤还借助了医生标注。可能的用途包括切片之间的查看、通过场的导数测量运动、紧凑存储，以及把不同设备的扫描对齐到同一坐标系中。

16. How do you know your numbers are correct?

The baselines are computed by the same code on the same hidden data, there are unit tests for the metrics and baselines, every run saves its settings and versions, and the final numbers were recomputed independently from the saved networks and matched to the second decimal.

> 基线方法由同一套代码在同样的隐藏数据上计算，指标和基线有单元测试，每次运行都保存设置和版本，最终数字还从保存的网络独立重新计算，结果精确到小数点后两位一致。

17. Did you choose the time scale on the test frames?

We first explored it on the hidden frames, so we repeated the choice on separate validation frames that were removed from training and never scored as test. The validation picked the same value, 0.1 (24.0 dB against 23.7 for 0.05 and 20.7 for 0.25), and all 14 new patients used it without any tuning.

> 我们最初是在隐藏帧上探索的，所以又在单独的验证帧上重新做了选择：这些帧从训练中移除，也从不作为测试评分。验证选出了同样的值0.1（24.0 dB，而0.05为23.7，0.25为20.7），14个新病人都直接使用它，没有任何调整。

18. Why not ReLU with Fourier features? That is the usual fix for ReLU.

We tested it with the frequency scale chosen on validation frames. Fourier features help ReLU a lot, from 22.3 to 26.6 dB, but SIREN is still 2.3 dB better with fewer weights, and it also has a lower temporal error.

> 我们测试了它，频率尺度在验证帧上选择。Fourier features对ReLU帮助很大，从22.3提高到26.6 dB，但SIREN在权重更少的情况下仍然好2.3 dB，时间误差也更低。

19. Did you try to make the network beat interpolation by itself?

Yes. Higher-order interpolation (cubic splines) is worse than linear, because edges of the wall move and the intensity jumps. Only motion-aware methods win: optical flow beats linear interpolation by 0.3 dB. A simple motion-based field, one template plus a displacement, did not reach plain SIREN; the heart wall thickens and moves through the slices. Motion-aware fields and priors learned over many patients are our next step.

> 试过。高阶插值（三次样条）比线性插值更差，因为心壁边缘在移动，亮度是跳变的。只有考虑运动的方法更好：光流比线性插值高0.3 dB。一个简单的运动场——一个模板加位移——没有达到普通SIREN的水平：心壁会变厚，并且会穿过切片移动。能建模运动的神经场以及从许多病人学到的先验是我们的下一步。

20. Is anything leaking from the test into training?

We checked the usual sources. Intensity scaling: the min and max of the training frames alone are identical to those of the whole scan, so nothing changes. Settings: chosen on validation frames. Train PSNR: computed exactly over all training voxels. The network never sees labels; only the 3D step and the heart mask for scoring use them, and we say so.

> 我们检查了常见的来源。亮度归一化：只用训练帧计算的最小值和最大值与整个扫描完全相同，没有任何变化。设置：在验证帧上选择。训练PSNR：在全部训练体素上精确计算。网络从不看到标注；只有三维步骤和用于评分的心脏掩码使用标注，我们也明确说明了这一点。